

Dense Expands, Sparse Anchors: Channel-Asymmetric Query Expansion for Hybrid Retrieval

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Abstract

LLM-based query expansion improves retrieval by generating document-like evidence. In hybrid retrieval, however, most evaluations fuse fixed top- L dense and sparse rankings. Because the cutoff controls both which cross-channel contributions enter fusion and how much of each ranking is accessed, gains measured at one L can change or reverse at another. We separate these effects by evaluating retrieval effectiveness under complete-list fusion and recording the policy-specific per-channel replay stopping depths at which its ordered top- K is certified. We then introduce DESA (Dense Expansion and Sparse Anchoring), a channel-asymmetric query expansion method. An LLM generates complementary reference passages; orthogonal residual expansion adds their new semantic directions to the dense query, while score-product anchoring incorporates their lexical evidence into sparse retrieval without broadening the original query’s lexical support. Across seven benchmarks, DESA improves nDCG@10 and Recall@20 over the unexpanded query by 3.82% and 2.38%, while reducing dense and sparse access depths by 36.90% and 36.56%. With equal dataset weighting, 63.31% of queries become shallower in both channels. However, both depths increase with Contriever on Touché-2020. These results support channel-specific integration of generated evidence and joint evaluation of retrieval effectiveness and access depth.

1 Introduction

Existing LLM-based query expansion methods augment the original query with generated content and typically evaluate retrieval gains at a preset top- L cutoff (Gao et al., 2023; Wang et al., 2023; Zhang et al., 2024; Liu and Zhang, 2025). In hybrid retrieval, however, top- L jointly determines how much of each ranking is read and which documents contribute to fusion; changing L may there-

(a) Dense expansion recovers missing semantics

Query A

How are various types of income taxed differently in the USA?

Generated references

Capital gains are taxed separately from **ordinary income**.
Long-term capital gains depend on the holding period.

Relevant passage

Rank 271 → 5

... **Long-term capital gains** are taxed below the marginal rate and differ from **wage or ordinary income** ...

Generated evidence supplies missing document language.

(b) Sparse anchoring preserves query evidence

Query B

How to find **cheaper alternatives** to a traditional **home telephone line**?

Generated references

Consider **VoIP services** as a home-phone alternative.
Some **internet service providers** offer integrated phone service.

Relevant passage

Rank 31 → 1

Use a **VOIP service provider** as an **alternative** to your **home phone line**.

Expansion-only match

Rank 5 → filtered

Internet Phone: VoIP dialer, reseller packages, login credentials, prices, and contact details ...

Green: generated | underline: original | Red: expansion-only

Figure 1: FiQA examples of DESA. Dense expansion recovers missing document semantics; sparse anchoring promotes jointly supported passages and excludes expansion-only matches. Ranks are channel ranks.

fore change the fused result itself (Zhang, 2026). A gain measured at a single cutoff is thus conditional on L and does not isolate the effect of query expansion.

To separate these factors, we first determine the fused result of each query construction from the complete dense and sparse rankings (Zhang, 2026). A fixed sound replay procedure then reveals the two rankings progressively and records the per-channel depths at which the same ordered top- K is

certified. This separates two questions: what fused result does an expansion produce, and how much ranked evidence is required to determine it? The answers need not move together. An expansion may improve retrieval effectiveness while pushing decisive evidence deeper into one channel. We therefore ask whether query expansion can improve effectiveness while reducing the access depth required from both channels.

Expansion effects are retriever-dependent (Weller et al., 2024; Li et al., 2026). On NFCorpus, applying the same expansion to both channels improves the fused result and reduces dense access depth, but increases sparse access depth. Generated terms substantially enlarge the set of documents with nonzero sparse scores, preventing the sparse ranking from exhausting as early as it does for the original query. This reversal motivates an asymmetric design: dense retrieval should absorb new semantic evidence, whereas sparse retrieval should restrict it to documents already supported by the original query.

We address this asymmetry with DESA (Dense Expansion and Sparse Anchoring), a channel-asymmetric query expansion method for hybrid retrieval. Given an original query, the LLM generates several complementary reference passages around the same information need. For dense retrieval, orthogonal residual expansion preserves the original query direction and adds only the new semantic component supplied by the references. For sparse retrieval, score-product anchoring multiplies each document’s scores under the original and expanded queries. Documents matched only by generated terms receive no anchored score, while the new terms may still reorder documents supported by the original query. Together, these operations broaden dense semantic coverage without broadening sparse support: Dense expands; sparse anchors. Figure 1 illustrates this division of labor through two retrieval examples.

Across seven public retrieval benchmarks, DESA improves equal-dataset macro nDCG@10 and Recall@20 over the unexpanded query by 3.82% and 2.38%, while reducing dense and sparse exact-access depths by 36.90% and 36.56%, respectively. With equal dataset weighting, 63.31% of queries become shallower in both channels. Mechanism experiments show that sparse anchoring contributes most of the effectiveness gain, while the dense orthogonal residual further reduces the ranked evidence required to determine the fused

result.

2 Related Work

2.1 LLM-Based Query Expansion

LLM-based query expansion supplements short queries with document-like language that may be absent from the original formulation. HyDE encodes a generated hypothetical document for dense retrieval, whereas Query2doc appends a pseudo-relevant document to the query for sparse and dense retrieval (Gao et al., 2023; Wang et al., 2023). Later methods generate different forms of evidence or sample multiple references to cover alternative expressions of the same information need (Jagerman et al., 2023; Zhang et al., 2024). Although generated text can supply useful lexical and semantic cues, its effect varies across queries, datasets, and retrievers, and irrelevant content may reduce retrieval effectiveness (Weller et al., 2024).

2.2 Evaluating Expansion under Hybrid Fusion

Query-expansion methods commonly evaluate retrieval using a fixed top- L prefix from each ranking (Zhang et al., 2024; Liu and Zhang, 2025). In hybrid retrieval, L also determines which cross-channel contributions enter fusion, making the measured expansion effect conditional on the selected cutoff. Following EAHR, we evaluate complete-list retrieval quality separately from channel access depth (Zhang, 2026), and ask whether query expansion can improve the former while reducing the latter.

2.3 Channel-Aware Integration

Generated evidence can enter retrieval through query text, dense representations, term weights, or rank fusion. Query2doc modifies the query text, MuGI pools representations from multiple references, Word2Passage reweights generated terms, and Exp4Fuse combines original and expanded rankings (Wang et al., 2023; Zhang et al., 2024; Choi et al., 2025; Liu and Zhang, 2025). DVCQR further constructs separate lexical and semantic views for sparse and dense retrieval (Li et al., 2026). These methods differ in where expansion is integrated; DESA instead focuses on how the same evidence should be constrained differently across the two channels.

QuDAR adaptively fuses four rankings obtained from original and expanded queries under sparse

and dense retrieval (Kim et al., 2026). DESA instead modifies the two rankings before fusion: generated evidence expands Dense semantics while Sparse remains anchored to the original query’s lexical support.

3 DESA

3.1 Method Overview

Given an original query q_o , DESA generates a set of complementary reference passages $R = \{r_1, \dots, r_n\}$. Dense retrieval keeps the original query as its base direction and adds only the new semantic component supplied by the references. Sparse retrieval combines the original and expanded scores so that generated terms can reorder documents already matched by the original query without introducing expansion-only matches. The resulting Dense and Sparse rankings are fused into the final result. Figure 2 summarizes this design: Dense expands; Sparse anchors.

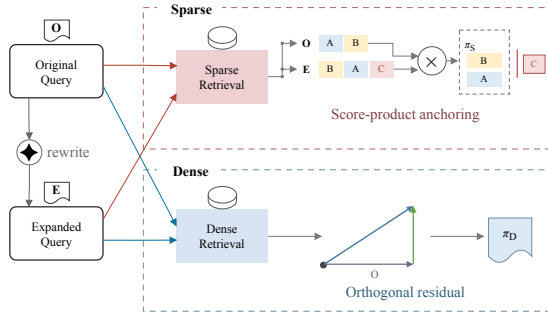


Figure 2: Design principle of DESA. The same generated evidence expands Dense semantics but remains anchored to the original query’s lexical support in Sparse retrieval.

3.2 Complementary Evidence Generation

The generator is prompted to produce n short passages that could plausibly occur in relevant documents. Each passage introduces a different alias, technical term, or paraphrase while retaining the entities, relations, numbers, polarity, temporal conditions, scope, and other constraints of q_o . The prompt prohibits broader information needs and invented answers. The passages therefore vary in expression rather than intent, allowing the two retrieval channels to draw additional evidence from a shared expansion. The complete prompt and decoding settings are provided in Appendix A.

3.3 Channel-Asymmetric Query Construction

3.3.1 Orthogonal Residual Expansion

Let f_D be the Dense encoder. We encode the original query and each query–reference pair as unit vectors:

$$\begin{aligned} \mathbf{v}_o &= \text{norm}(f_D(q_o)), \\ \mathbf{v}_i &= \text{norm}(f_D(\text{concat}(q_o, r_i))), \\ \bar{\mathbf{v}}_r &= \frac{1}{n} \sum_{i=1}^n \mathbf{v}_i. \end{aligned}$$

The component of $\bar{\mathbf{v}}_r$ parallel to the original query is removed, and the remaining residual is added to $\mathbf{v}_o$:

$$\mathbf{z} = \bar{\mathbf{v}}_r - (\mathbf{v}_o^\top \bar{\mathbf{v}}_r) \mathbf{v}_o, \quad \mathbf{e}_D = \text{norm}(\mathbf{v}_o + \mathbf{z}).$$

Documents are ranked by their similarity to $\mathbf{e}_D$, producing the Dense ranking π_D . Because $\mathbf{z} \perp \mathbf{v}_o$ and $\|\mathbf{z}\| \leq 1$,

$$\angle(\mathbf{e}_D, \mathbf{v}_o) = \arctan \|\mathbf{z}\| \leq \frac{\pi}{4}.$$

The references can therefore add semantic directions absent from the original representation while keeping the expanded query within 45° of it. If they contribute no orthogonal component, then $\mathbf{e}_D = \mathbf{v}_o$.

3.3.2 Score-Product Anchoring

For Sparse retrieval, the expanded query concatenates the original query and all reference passages:

$$q_r = \text{concat}(q_o, r_1, \dots, r_n).$$

Let $s_S(q, d)$ be a non-negative additive Sparse scorer. Each document d receives an original score, an expanded score, and their product:

$$\begin{aligned} s_o(d) &= s_S(q_o, d), & s_r(d) &= s_S(q_r, d), \\ s_A(d) &= s_o(d) s_r(d). \end{aligned}$$

Because q_r contains q_o , every document supported by the original query is also supported by the expanded query. Hence,

$$\text{supp}(s_A) = \text{supp}(s_o) \cap \text{supp}(s_r) = \text{supp}(s_o).$$

Sorting the positive anchored scores produces the Sparse ranking π_S . Generated terms may promote or demote documents within the original support, but documents matched only by the expansion

receive zero anchored score and are excluded. If s_r is merely a positive multiple of s_o , the original Sparse order is unchanged; positive global score rescaling also leaves the ranking invariant.

3.4 Complete Fusion and Access Measurement

We fuse the complete Dense and Sparse rankings using equal-weight reciprocal rank fusion (Cormack et al., 2009):

$$F(d) = \sum_{c \in \{D, S\}} \frac{1}{\kappa + \text{rank}_{\pi_c}(d)}.$$

A document absent from a channel contributes zero. Sorting by $F(d)$, with document identifiers breaking ties, defines the ordered target T_K . Retrieval effectiveness is evaluated directly on this complete-list result.

To measure channel access, a fixed replay procedure progressively reveals π_D and π_S . After L_c items have been read from channel c , the largest possible contribution from its next unread item is

$$u_c(L_c) = \frac{1}{\kappa + L_c + 1}.$$

The replay reads from the channel with the larger bound, breaking ties in favor of Dense, until the membership and order of T_K can no longer be changed by unread documents. It then records the Dense and Sparse depths (L_D, L_S), which quantify how much ranked evidence each channel must reveal to determine the complete-list result.

4 Experiments

We first examine whether a fixed Top- L cutoff changes the measured effect of query expansion. We then evaluate whether DESA improves retrieval quality while reducing the ranking depth required from Dense and Sparse retrieval. The remaining experiments separate the contributions of the two channel operators and test robustness across generators, dense encoders, reference counts, and corpus sizes.

4.1 Evaluation Protocol

To measure cutoff sensitivity, we vary $L \in \{10, 20, 50, 100, 200, 500, 1000\}$. At each cutoff, the Top- L Dense and Sparse rankings are fused using equal-weight RRF. We compute nDCG@10 and Recall@20, compare the ordering of query constructions across cutoff depths, and measure

whether each fixed- L result reproduces the ordered Top-20 from complete-list fusion (Järvelin and Kekäläinen, 2002; Cormack et al., 2009).

We also examine whether the cutoff changes the judgment of an expansion relative to the unexpanded query. For each query and generation draw, the metric difference between an expansion and Original is labeled as a gain, tie, or loss under both fixed- L and complete-list fusion. We report how often this label changes and separately count strict reversals, where a gain becomes a loss or vice versa.

All remaining experiments use the complete-list fusion defined in Section 3.4. Retrieval quality is measured by nDCG@10 and Recall@20. Channel access is measured by the Dense and Sparse replay stopping depths (L_D, L_S). We report their changes from Original and the proportion of queries for which both depths decrease. Sparse support size and exhaustion rate are used to explain changes in lexical access.

4.2 Datasets and Retrieval Setup

We evaluate on seven English BEIR collections: SciFact (Wadden et al., 2020), NFCorpus (Boteva et al., 2016), TREC-COVID (Voorhees et al., 2020), FiQA (Maia et al., 2018), ArguAna (Wachsmuth et al., 2018), Touché-2020 (Bondarenko et al., 2020), and SCIDOCS (Cohan et al., 2020). These datasets cover scientific claim verification, biomedical and financial retrieval, argument retrieval, and scientific document recommendation. Unless otherwise stated, the cutoff, controlled, and mechanism analyses use all seven datasets, with each dataset receiving equal weight in macro averages.

For the scale analysis, we construct nested TREC-COVID corpora containing 25,000, 50,000, 100,000, and all documents. Every snapshot retains the judged relevant documents, while the remaining documents are added in a fixed order. Each smaller corpus is therefore a strict subset of the larger ones.

The primary generator is Qwen2.5-7B-Instruct (Qwen Team, 2024). For each query, we perform three generation draws and produce five complementary reference passages per draw. The generator robustness experiment uses Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) on up to 100 queries selected by a fixed hash from FiQA, ArguAna, Touché-2020, and SCIDOCS.

Dense retrieval uses BGE-small-en-v1.5 (Xiao et al., 2023), with normalized query and document

vectors ranked by dot product. Sparse retrieval uses Lucene BM25 through Pyserini 2.3.0, with $k_1 = 0.9$ and $b = 0.4$ (Robertson and Zaragoza, 2009; Lin et al., 2021). Contriever (Izacard et al., 2022) is used as a second Dense encoder. Score ties are broken by document identifier.

4.3 Comparison Methods

The controlled comparison includes Original and Shared expansion. Original submits the unexpanded query to both retrieval channels. Shared expansion uses the same generated references as DESA but applies them directly to both channels: Dense retrieval averages the representations of the five query and reference pairs, while Sparse retrieval concatenates the original query and all references. The comparison between Shared expansion and DESA therefore tests whether Dense and Sparse benefit from different integration rules.

We also compare with three representative generative expansion methods. HyDE encodes generated hypothetical documents for Dense retrieval (Gao et al., 2023). Query2doc appends a generated pseudo-document to the query (Wang et al., 2023). MuGI generates multiple references, pools their contextualized Dense representations, and uses query-weighted concatenation for Sparse retrieval (Zhang et al., 2024). Since all three reproduced pipelines are available on FiQA, ArguAna, Touché-2020, and SCIDOCS, the full-method comparison uses this common four-dataset subset.

We further compare with the rank-based QuDAR-simple variant, which uses the same generated references and fuses the Top-1000 OS, ES, OD, and ED rankings with RRF ($k = 60$) (Kim et al., 2026).

4.4 Mechanism and Robustness Analysis

We isolate the two channel operators with matched controls. For Dense retrieval, the original Sparse ranking is paired with either unprojected contextual expansion or orthogonal residual expansion. For Sparse retrieval, the original Dense ranking is paired with either direct rewriting or score-product anchoring. A 2×2 comparison further evaluates Original, each proposed operator alone, and their combination.

We then examine how each operator changes its retrieval channel. Dense behavior is measured by the angle between the original and expanded query vectors. Sparse behavior is measured by Top-20 turnover and by whether relevant documents move

upward in the ranking. We compare these quantities with the corresponding changes in retrieval quality and access depth using within-dataset quartiles and Spearman correlations.

Additional experiments vary the number of reference passages over $\{1, 3, 5\}$. Score-product anchoring is compared with a binary mask that retains expanded-query scores only for documents matched by the original query, separating support preservation from the use of graded original scores. We also test Sparse retrieval using only the generated passages. Robustness experiments vary the RRF constant over $\{2, 20, 60, 100\}$, replace the generator and Dense encoder, and evaluate the nested TREC-COVID corpora.

4.5 Statistical Analysis

The three generation draws are averaged within each query before aggregation. We report results for each dataset and equal-dataset macro averages. The 95% confidence intervals are obtained from 10,000 stratified bootstrap samples, with queries resampled within each dataset.

The prespecified primary comparisons are DESA against Original and DESA against Shared expansion. We use 10,000 stratified paired sign-flip randomizations and apply Holm correction across $nDCG@10$, $Recall@20$, L_D , and L_S separately for each comparator. The operator controls and the matched QuDAR comparison use the same draw-averaged, query-level paired procedure, with Holm correction across $nDCG@10$ and $Recall@20$.

5 Results

5.1 Fixed Cutoffs Alter Measured Effects

Figure 3 compares the query-level judgment of DESA relative to Original at $L = 50$ with the judgment under complete-list fusion. Each query-draw pair is labeled as a gain, tie, or loss from the corresponding metric difference. Diagonal cells preserve that judgment; off-diagonal cells show cases in which the fixed cutoff changes it.

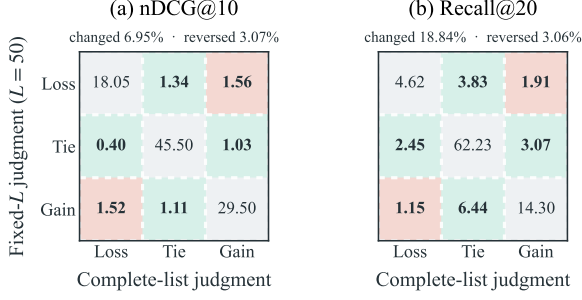


Figure 3: Transition between the DESA-versus-Original judgment at fixed $L = 50$ (rows) and under complete-list fusion (columns). Cells report the equal-dataset percentage of query-draw pairs. Green off-diagonal cells change among gain, tie, and loss; red cells are strict gain-loss reversals.

The fixed cutoff changes 6.95% of nDCG judgments and 18.84% of Recall judgments. Strict gain-loss reversals account for 3.07% and 3.06%, respectively. Thus, a practical fixed cutoff can change whether expansion is judged beneficial for an individual query, rather than merely perturbing the ranking. Results for other cutoffs, complete macro curves, and ordered Top-20 agreement are reported in Appendix Tables 6 and 7.

5.2 Retrieval Effectiveness and Access Depth

Under complete-list fusion, DESA obtains the highest macro nDCG@10 and Recall@20 in the seven-dataset controlled comparison, reaching .4747 and .4695. Relative to Original, these values correspond to gains of 3.82% and 2.38%. The equal-dataset averages of the within-dataset depth reductions are 36.90% for Dense and 36.56% for Sparse. The mean proportion of queries that become shallower in both channels is 63.31%. All four improvements remain significant after Holm correction ($p_{\text{Holm}} = .0004$).

Method	nDCG@10	Recall@20	L_D	L_S
<i>No expansion</i>				
Original	0.4572	0.4586	1752.4	1455.1
<i>Symmetric expansion</i>				
Shared	0.4671	0.4666	607.4	606.7
<i>Single-channel construction</i>				
Dense only	0.4621	0.4637	1386.5	1219.9
Sparse only	<u>0.4714</u>	<u>0.4678</u>	884.4	822.0
<i>Channel-asymmetric construction</i>				
DESA	0.4747	0.4695	<u>728.9</u>	<u>673.2</u>

Table 1: Equal-dataset macro results for the seven-dataset controlled comparison. Higher is better for retrieval metrics; lower is better for replay stopping depths. Raw depths are averaged across datasets; percentage reductions in the text are computed within each dataset before macro-averaging.

Compared with Shared expansion, DESA improves macro nDCG by 1.61% and Recall by 0.61%. The absolute nDCG difference is +.0075 and remains significant after correction ($p_{\text{Holm}} = .0012$). The Recall difference is +.0028 ($p_{\text{Holm}} = .2736$), while the differences in Dense and Sparse depth are also nonsignificant. DESA therefore provides a significant nDCG advantage over Shared expansion.

NFCorpus illustrates the effect of Sparse anchoring. Shared expansion increases the mean Sparse support from 561.5 to 3211.6 documents and raises Sparse stopping depth by 105.82% relative to Original. DESA preserves the original support size and reduces Sparse depth by 23.15%, while also reducing Dense depth by 15.03%. Anchoring allows generated terms to improve the ordering of originally matched documents without expanding the lexical support.

5.3 Contributions of the Channel Operators

The 2×2 experiment shows that the two channel operators are complementary. Dense residual expansion alone improves nDCG and Recall by 1.08% and 1.11%, with Dense and Sparse depth reductions of 8.16% and 7.49%. Sparse anchoring alone produces larger quality gains of 3.11% and 2.01%, together with depth reductions of 28.77% and 27.74%. Combining both operators gives the full DESA gains of 3.82% and 2.38%, while reducing Dense and Sparse depth by 36.90% and 36.56%.

Sparse anchoring contributes most of the quality gain, while Dense residual expansion further reduces access depth. Their isolated controls produce only small differences, whereas the complete channel-asymmetric construction significantly outperforms Shared expansion in nDCG. DESA’s advantage therefore comes from combining the two channel-specific operations.

The observed rankings confirm the intended channel behavior (Figure 4). Across 11,327 query and generation-draw pairs, the mean Dense angle is 11.98° , and the maximum is 27.87° , below the analytic 45° bound. Sparse anchoring changes 26.15% of the original Sparse Top-20 on average.

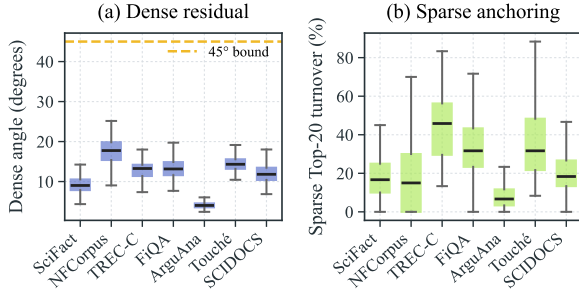


Figure 4: Realized behavior of the two channel operators over three generation draws on seven datasets. Dense angles are measured from the original query; Sparse turnover is one minus Top-20 overlap with the original sparse ranking. Boxes show the interquartile range and median; whiskers extend to 1.5 times the interquartile range, with outliers omitted.

Relevant-document reciprocal-rank gain correlates more strongly with Sparse-only nDCG gain than overall Top-20 turnover does (.391 vs. .078). Sparse anchoring therefore improves retrieval by promoting relevant documents within the original support, while Dense residual expansion adds semantic evidence without excessive drift.

Additional binary-mask, reference-count, and operator-control results are reported in Appendix C.

5.4 Comparison with Recent and Prior Methods

Under matched generated evidence, QuDAR-simple RRF obtains .4761 nDCG and .4714 Recall, compared with .4747 and .4695 for DESA. Neither paired difference is significant after correction. QuDAR-simple fuses four Top-1000 rankings, corresponding to 4,000 rank entries per query. DESA requires a mean of 1,402.1 entries to certify its fused result under the replay policy, a reduction of 64.95%. DESA therefore matches the retrieval quality of the rank-based QuDAR variant with substantially less fusion-side ranking access.

On the common four-dataset subset, DESA improves nDCG and Recall over HyDE by 16.58% and 10.78%, while reducing both stopping depths by about 91%. Relative to MuGI, the quality gains are 5.58% and 2.92%, and Dense and Sparse depths decrease by 50.22% and 51.77%. DESA therefore outperforms both methods in retrieval quality and ranking access under the shared evaluation protocol.

Query2doc has 0.45% higher nDCG, whereas DESA has 0.57% higher Recall. DESA also reduces Dense and Sparse stopping depths by 18.90%

Method	nDCG@10	Recall@20	L_D	L_S
No expansion	.3325	.5142	1278.2	1228.4
HyDE	.2956	.4746	7918.3	7536.6
MuGI	.3264	.5109	1409.0	1406.3
Query2doc	.3461	.5229	865.0	857.8
DESA	.3446	.5258	701.5	678.2

Table 2: Equal-dataset macro comparison with prior query expansion methods on the common four-dataset subset.

and 20.93%. The two methods have similar retrieval quality, but DESA requires less ranking evidence to determine the fused result.

5.5 Robustness and Failure Boundary

DESA retains its quality gains across the tested generator, Dense encoder, and corpus sizes. Access depth decreases in most settings, with Contriever on Touché-2020 as the only observed failure case. Detailed generator, encoder, and corpus-scale results are reported in Appendix C.4.

6 Conclusion

A fixed top- L cutoff limits how much of each channel ranking enters fusion and can change the measured gains and ordering of query expansion methods. We instead evaluate retrieval effectiveness with complete-list fusion and record the policy-specific per-channel replay depths at which the fused result is certified. DESA uses an LLM to generate complementary reference passages for channel-asymmetric query expansion. Orthogonal residual expansion adds their new semantic directions to the dense query, while score-product anchoring incorporates their lexical evidence through a document-wise score product without broadening the original query’s lexical support.

Across seven evaluation datasets, DESA improves nDCG@10 and Recall@20 over the unexpanded query by 3.82% and 2.38%, with dense and sparse access-depth reductions of 36.90% and 36.56%. With datasets weighted equally, 63.31% of queries become shallower in both channels. This access benefit is not universal: depths increase with Contriever on Touché-2020, and the reductions are non-monotonic across corpus sizes. Query expansion for hybrid retrieval should therefore integrate generated evidence according to each channel’s matching mechanism and evaluate retrieval effectiveness together with access depth.

7 Limitations

Our experiments cover seven English BEIR collections, two instruction-tuned generators, two dense encoders, and BM25 as the sparse retriever. The results therefore do not establish the same behavior for other languages, learned sparse retrievers, or other fusion rules. Score-product anchoring additionally assumes a non-negative sparse scoring function, and the access analysis fixes equally weighted WRRF, $K = 20$, and a specific incremental replay policy.

The reported access measurements are logical certification depths obtained by replaying stored rankings. They are not wall-clock latency and do not include reference generation, query encoding, indexing, or communication costs. An online system would need an incremental retrieval interface before a reduction in replay depth could be interpreted as an end-to-end speedup.

The direct QuDAR comparison is limited to QuDAR-simple RRF. It follows the official four-signal, Top-1000 rank-fusion configuration and reuses the same generated evidence, but our frozen ranking artifacts do not contain the normalized retrieval scores required by QuDAR-confidence, and we do not add new LLM relevance judgments for QuDAR-llm. The comparison therefore supports parity with the reconstructable rank-based variant, not with every QuDAR weighting strategy.

Generated references may contain irrelevant or unsupported expressions despite the intent-preserving prompt and structured validation. We average three draws and report parser fallback behavior, but this does not cover the full variation of possible generators and decoding settings. Finally, the benchmark relevance judgments are incomplete, as is common in pooled retrieval collections; methods that retrieve documents outside the original pools may therefore be underestimated. Our MuGI comparison reproduces its generation and integration components without pseudo-relevance-feedback calibration, and the complete prior-method comparison is limited to the four datasets on which all reproduced pipelines were available.

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useful aliases, technical terminology, and paraphrases, but do not broaden the information need or invent a specific answer. Each passage must be one sentence of at most 25 words.

Return one single-line JSON array containing exactly five distinct strings. Count the five array items before responding. Do not output an object key, query identifier, Markdown, placeholders, or any other text. Every item must contain a substantive passage.

The query is supplied in a canonical JSON object with a single query field. XGrammar 0.2.1 constrains the response to a JSON array containing exactly five strings. Validation rejects empty passages, placeholders, and duplicates. An invalid response is retried once with the same seed, the previous response, and validation feedback. If the retry also fails, the failure is recorded and the corresponding query and generation run fall back to the unexpanded query.

The second generator is Mistral-7B-Instruct-v0.3 at revision

c170c708c41dac9275d15a8fff4eca08d52bab71.

It uses the same decoding, batching, and validation settings.

A.2 Query Construction

All averaging, projection, and normalization operations in orthogonal residual expansion use float32. Shared expansion uses the same representations of the query-reference pairs, but averages and normalizes them without removing the component parallel to the original query. Each pair is encoded as query [SEP] reference.

For sparse retrieval, the expanded query contains one copy of the original query followed by the selected reference passages. BM25 scores under the original and expanded queries are stored separately before applying the document-wise score product. The binary-mask ablation retains the expanded-query score for documents matched by both queries. The references-only ablation removes the original query before retrieval. Our construction neither repeats the original query nor uses the MuGI repetition coefficient.

A.3 Baseline Reproduction

All generative baselines use the same primary generator so that their differences mainly reflect query construction rather than model capacity. The exact prompts are included in the accompanying repository.

HyDE generates eight hypothetical texts per run, encodes them independently as documents, and averages their normalized representations. Its sparse channel uses the original query. FiQA, ArguAna, SciFact, and TREC-COVID use task-specific generation instructions; the remaining datasets use a general retrieval instruction. HyDE allows up to 512 new tokens per generation.

Query2doc generates one pseudo-relevant document of at most 120 words. Dense retrieval encodes the concatenation of the original query and pseudo-document. Sparse retrieval concatenates five copies of the original query with the pseudo-document.

MuGI generates five pseudo-references. Dense retrieval averages the representations obtained by concatenating the original query with each reference. Sparse retrieval repeats the original query

$$\left\lfloor \frac{|R|}{|q_o|\beta} \right\rfloor$$

times before appending all references, where $|\cdot|$ denotes character length and $\beta = 4$. We reproduce MuGI’s generation and query integration components without its pseudo-relevance feedback calibration.

B Operator Properties and Mechanism Diagnostics

B.1 Properties of the Channel Operators

For the Dense construction, each contextual vector $\mathbf{v}_i$ is unit normalized. The triangle inequality gives

$$\|\bar{\mathbf{v}}_r\| \leq \frac{1}{n} \sum_{i=1}^n \|\mathbf{v}_i\| = 1.$$

Orthogonal projection cannot increase the norm, so $\|\mathbf{z}\| \leq 1$. Since $\mathbf{v}_o^\top \mathbf{z} = 0$ and $\|\mathbf{v}_o\| = 1$,

$$\mathbf{e}_D^\top \mathbf{v}_o = \frac{1}{\sqrt{1 + \|\mathbf{z}\|^2}},$$

$$\angle(\mathbf{e}_D, \mathbf{v}_o) = \arctan \|\mathbf{z}\| \leq \frac{\pi}{4}.$$

If $\mathbf{z} = \mathbf{0}$, normalization leaves $\mathbf{v}_o$ unchanged.

For Sparse, non-negativity gives $\text{supp}(s_o s_r) = \text{supp}(s_o) \cap \text{supp}(s_r)$. Additivity and the inclusion of q_o in q_r imply $\text{supp}(s_o) \subseteq \text{supp}(s_r)$, yielding exact support preservation. If $s_r = c s_o$ for $c > 0$, then $s_A = c s_o^2$; squaring is strictly increasing on positive scores, so the ranking is unchanged. Multiplying s_o or s_r by a positive global constant only rescales every anchored score and likewise leaves the ranking unchanged.

B.2 Mechanism Diagnostics

Table 3 reports the quantities used to inspect the realized operators. Dense angle is measured between the original and expanded query vectors. Sparse turnover is one minus Top-20 overlap with the original sparse ranking. Relevant RR change averages the difference in reciprocal rank for judged relevant documents that appear in the exported rankings. All macro values give each dataset equal weight.

The within-dataset quartiles in Figure 5 are not monotonic. Larger Dense angles do not consistently improve either nDCG or access depth, and greater Sparse turnover does not consistently do so either. The descriptive correlations lead to the same distinction: relevant-document reciprocal-rank gain has a mean Spearman correlation of .391 with Sparse-only nDCG gain, whereas overall turnover has a correlation of .078. The corresponding Dense-angle correlations with nDCG gain and depth reduction are .010 and .011. These observations do not support an angle or turnover threshold.

The ideal support-preservation property assumes a non-negative additive Sparse scorer and complete score export. It holds exactly in the exported rankings for six datasets. On Touché-2020, 23 of 147 query-draw pairs omit one or two extremely low-scoring tail documents, for 25 omissions in total. Mean support retention is 99.99990%, and the minimum is 99.99881%. We retain these cases in the diagnostic artifacts and do not treat the practical Lucene export as exactly support preserving on this dataset.

C Retrieval and Access Analysis

C.1 Per-Dataset Controlled Results

Table 4 exposes the dataset-level values behind the controlled macro comparison. The three methods use identical generated references where applicable. Sparse support is the mean number of positive-score documents; exhaustion is the fraction of query-draw rankings whose finite Sparse support is exhausted during replay.

C.2 Primary Statistical Tests

Table 5 reports the complete seven-dataset statistical record. For retrieval metrics, the favorable difference is DESA minus the comparator; for depth, it is the comparator minus DESA, so positive values always favor DESA. Holm correction is applied over the four outcomes separately for each prespecified comparator.

C.3 Cutoff, Operator-Control, and QuDAR Details

Table 6 reports the complete macro nDCG@10 values behind the fixed-cutoff analysis.

Table 7 reports the query-draw diagnostic behind the fixed-cutoff discussion. Conclusion change compares the sign of the method-Original difference under fixed- L and complete-list fusion. Strict reversal excludes changes to or from a tie.

Table 8 isolates the transformation applied within each channel. The control pairs use identical generated references and keep the other channel at its original-query ranking. Effects average the three draws within query before paired inference.

C.4 Additional Ablations and Robustness

The binary-mask ablation retains the same Sparse support as score-product anchoring but discards the magnitude of the original-query score. Replacing the score product with this mask lowers macro nDCG@10 by 1.54% and Recall@20 by 0.74%. The graded original score therefore improves the ordering of documents within the retained support.

Reference count affects both effectiveness and access depth. Increasing the number of complementary passages from one to three improves nDCG@10 by 1.91% and reduces Dense and Sparse stopping depths by 25.46% and 26.61%. Increasing the count from three to five adds 0.35% nDCG@10 and 0.43% Recall@20, with further depth reductions of 6.05% and 5.62%.

With Mistral as the generator, DESA produces non-negative nDCG@10 and Recall@20 changes on every dataset in the sampled four-dataset subset. Relative to Original, Dense depth reductions range from 7.12% to 47.61%, and Sparse depth reductions range from 7.16% to 47.62%. On ArguAna, 10% of the 100 sampled queries fall back to Original after invalid structured output.

Replacing BGE with Contriever also improves both retrieval metrics on all four sampled datasets. Touché-2020 is the observed access-depth failure case: Dense and Sparse stopping depths increase by 14.29% and 4.34%, respectively.

Across the 25k, 50k, 100k, and 171k TREC-COVID corpora, nDCG@10 gains range from 3.69% to 5.78%, and Recall@20 gains range from 5.56% to 9.21%. The Dense and Sparse depth reductions are 68.12%/68.16%, 65.11%/65.14%, 79.83%/79.84%, and 74.89%/66.71%, respectively. The access benefit remains substantial at every

Dataset	Dense angle (°)	Sparse Top-20 turnover (%)	Relevant RR change	Support retained (%)
SciFact	9.38	18.88	.0211	100.0000
NFCorpus	17.65	18.69	.0147	100.0000
TREC-COVID	12.99	45.37	.0021	100.0000
FiQA	13.34	33.91	.0204	100.0000
ArguAna	4.02	8.14	.0048	100.0000
Touché-2020	14.40	37.07	.0091	99.9999
SCIDOCS	12.05	20.98	.0028	100.0000
Equal-dataset macro	11.98	26.15	.0107	100.0000

Table 3: Mechanism diagnostics computed from the frozen query constructions and rankings. Support retained is the mean per-query–draw fraction of the original sparse support; the displayed macro rounds to 100.0000%.

Dataset	Method	Queries	nDCG@10	Recall@20	L_D	L_S	Sparse support	Exhaust. (%)
ArguAna	Original	1406	.3941	.9203	157.1	156.7	8587.8	0.00
	Shared		.4004	.9265	123.4	123.0	8616.8	0.00
	DESA		.3991	.9237	133.1	132.7	8587.8	0.00
FiQA	Original	648	.3649	.5260	2179.0	2076.3	28384.6	1.23
	Shared		.3635	.5391	1104.1	1103.6	46673.3	0.00
	DESA		.3780	.5450	1096.2	1065.6	28384.6	0.93
NFCorpus	Original	323	.3670	.2099	474.4	173.7	561.5	48.61
	Shared		.3780	.2260	359.1	357.6	3211.6	1.14
	DESA		.3793	.2195	403.1	133.5	561.5	46.75
SCIDOCS	Original	1000	.1923	.2721	694.8	599.3	11503.0	2.70
	Shared		.1905	.2790	372.7	372.2	23239.6	0.00
	DESA		.1940	.2787	448.8	387.2	11503.0	1.63
SciFact	Original	300	.7253	.9036	409.1	368.6	2765.7	4.33
	Shared		.7325	.9182	284.1	282.6	4489.0	0.33
	DESA		.7427	.9200	318.3	291.6	2765.7	3.89
TREC-COVID	Original	50	.7781	.0398	6270.9	4729.9	92683.8	2.00
	Shared		.8169	.0433	787.3	786.9	141483.1	0.00
	DESA		.8224	.0435	1574.8	1574.4	92683.8	0.00
Touché-2020	Original	49	.3786	.3382	2081.8	2081.4	151619.0	0.00
	Shared		.3883	.3344	1221.5	1221.3	284677.0	0.00
	DESA		.4071	.3560	1127.8	1127.4	151618.9	0.00

Table 4: Per-dataset controlled results for the seven evaluation collections.

Comparison / outcome	Effect [95% CI]	p/p_{Holm}
Original / nDCG@10	.0175 [.0111, .0238]	.0001 / .0004
Original / Recall@20	.0109 [.0070, .0151]	.0001 / .0004
Original / L_D	1023.6 [377.5, 2012.6]	.0001 / .0004
Original / L_S	781.9 [356.1, 1339.7]	.0001 / .0004
Shared / nDCG@10	.0075 [.0036, .0116]	.0003 / .0012
Shared / Recall@20	.0028 [-.0003, .0061]	.0912 / .2736
Shared / L_D	-121.4 [-367.0, 39.7]	.3157 / .6313
Shared / L_S	-66.5 [-310.6, 93.0]	.7909 / .7909

Table 5: Stratified paired tests across the seven evaluation collections after averaging the three generation draws within query.

tested size and reaches its maximum at 100k documents.

C.5 Retrieval Implementation

BGE-small-en-v1.5 uses revision

5c38ec7c405ec4b44b94cc5a9bb96e735b38267a.

Queries are prefixed with “Represent this sentence for searching relevant passages:” and encoded with a maximum length of 512 tokens. Con-

triever uses mean pooling and revision

2bd46a25019aeea091fd42d1f0fd4801675cf699.

All query and document embeddings are normalized.

Pyserini 2.3.0 runs with Java 21 and uses DefaultEnglishAnalyzer. A separate Lucene index is built for each dataset and each TREC-COVID snapshot, so BM25 collection statistics are computed within the corresponding corpus. Documents with equal retrieval or fusion scores are ordered by the UTF-8 order of their identifiers.

C.6 Access-Depth Replay

The replay begins with empty Dense and Sparse prefixes. After l_c items have been read from channel c , the maximum contribution of its next unread document is

$$u_c(l_c) = \frac{1}{60 + l_c + 1}.$$

The bound is zero after a channel is exhausted.

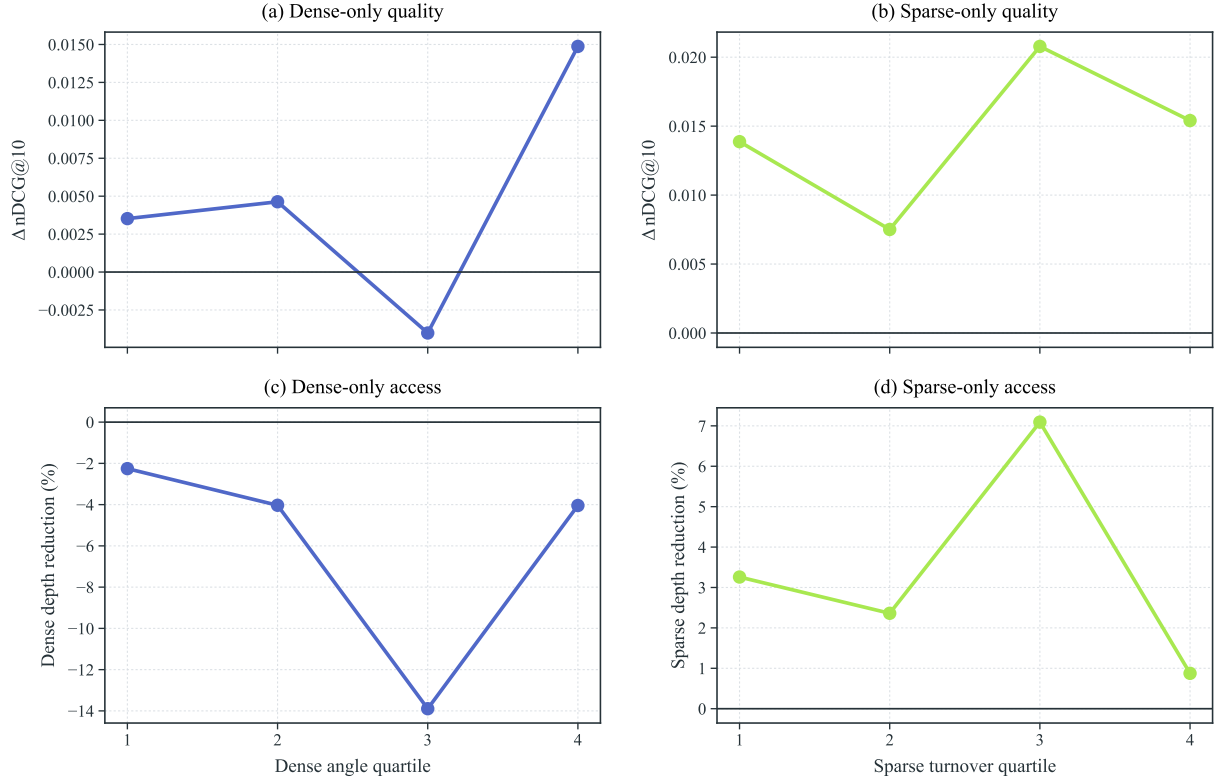


Figure 5: Equal-dataset means after assigning queries to within-dataset quartiles of Dense angle or Sparse Top-20 turnover. Positive depth reduction means shallower replay. The nonmonotonic curves are reported as descriptive mechanism diagnostics, not as a basis for threshold selection.

Method	$L = 10$	$L = 20$	$L = 50$	$L = 100$	$L = 200$	$L = 500$	$L = 1000$	Complete
No expansion	.4478	.4489	.4512	.4553	.4565	.4569	.4571	.4572
Shared	.4630	.4633	.4641	.4645	.4666	.4670	.4671	.4671
Dense only	.4566	.4568	.4582	.4608	.4617	.4622	.4621	.4621
Sparse only	.4661	.4676	.4654	.4686	.4704	.4713	.4715	.4714
MuGI-style (controlled)	.4612	.4615	.4608	.4613	.4630	.4636	.4637	.4638
Sparse binary mask	.4635	.4637	.4639	.4649	.4670	.4672	.4675	.4675
Reference-only sparse	.4534	.4537	.4529	.4537	.4555	.4559	.4564	.4565
DESA	.4738	.4727	.4709	.4727	.4740	.4743	.4747	.4747

Table 6: Equal-dataset macro nDCG@10 under fixed top- L and complete-list fusion on the seven evaluation datasets. Bold marks the maximum within each row.

L	Exact Top-20 (%)	nDCG (%) Change	nDCG (%) Reverse	Recall (%) Change	Recall (%) Reverse
20	1.92	18.23	8.84	22.50	3.84
50	16.54	6.95	3.07	18.84	3.06
100	47.40	2.17	0.73	11.74	0.99
1000	92.62	0.19	0.01	1.18	0.00

Table 7: DESA query-draw sensitivity to fixed Top- L , macro-averaged with equal dataset weight. Exact Top-20 compares with complete-list fusion; conclusion columns compare DESA with Original.

At each step, the replay reads from the channel with the larger next-unread bound, breaking ties in favor of Dense.

For an observed document, its accumulated contribution is a score lower bound. If the document

has not appeared in the other channel, that channel’s next-unread contribution supplies an additional upper bound. A document unseen in both channels is bounded by the sum of the two next-unread contributions.

The replay stops when these bounds certify both the membership and order of the current top-20. The stopping depths are recorded as $(L_D, L_S) = (l_D, l_S)$.

C.7 Artifact Traceability

Each generation record contains the raw response, parsed passages, prompt and model hashes, decoding parameters, seed, retry history, runtime lock, and code commit. Dense and Sparse rankings are stored by channel in a compressed ranking store

Construction	nDCG@10	Recall@20
Original	.4572	.4586
Contextual Dense	.4622	.4637
Residual Dense	.4621	.4637
Rewrite Sparse	.4686	.4670
Product Sparse	.4714	.4678
Paired effect (proposed – control)		
Dense residual, nDCG	–.00005 [–.00086, .00082], $p_H = 1.0$	
Dense residual, Recall	–.00006 [–.00043, .00033], $p_H = 1.0$	
Sparse product, nDCG	.00283 [–.00126, .00693], $p_H = .3650$	
Sparse product, Recall	.00078 [–.00184, .00331], $p_H = .5689$	

Table 8: Equal-dataset macro functional controls and query-level paired effects. Holm correction is over nDCG and Recall within each control pair.

Dataset	nDCG@10		Recall@20	
	DESA	QuDAR	DESA	QuDAR
ArguAna	.3991	.3985	.9237	.9253
FiQA	.3780	.3833	.5450	.5554
NFCorpus	.3793	.3780	.2195	.2227
SCIDOCs	.1940	.1957	.2787	.2811
SciFact	.7427	.7403	.9200	.9211
TREC-COVID	.8224	.8233	.0435	.0438
Touché-2020	.4071	.4137	.3560	.3500
Equal-dataset macro	.4747	.4761	.4695	.4714

Table 9: Matched-evidence comparison with QuDAR-simple RRF. QuDAR fuses the Top-1000 OS, ES, OD, and ED rankings with RRF constant 60. DESA–QuDAR paired effects are $-.00145$ [$-.00478, .00181$] for nDCG and $-.00188$ [$-.00400, .00019$] for Recall, with $p_H = .3966$ and $.1668$.

and assigned SHA-256 hashes. Each query-level result contains the method, generation run, effectiveness metrics, (L_D, L_S) , sparse support and exhaustion status, ordered top-20, and hashes of the generation, rankings, method configuration, and replay trace.

The Python environment is fixed by `uv.lock`, and the Java version is checked before retrieval. Before final evaluation, the code, configurations, prompts, source inputs, model artifacts, generations, and ranking stores are recorded in a hash manifest. All reported tables are reconstructed from the resulting query-level records.